

# Node Structure Visualization

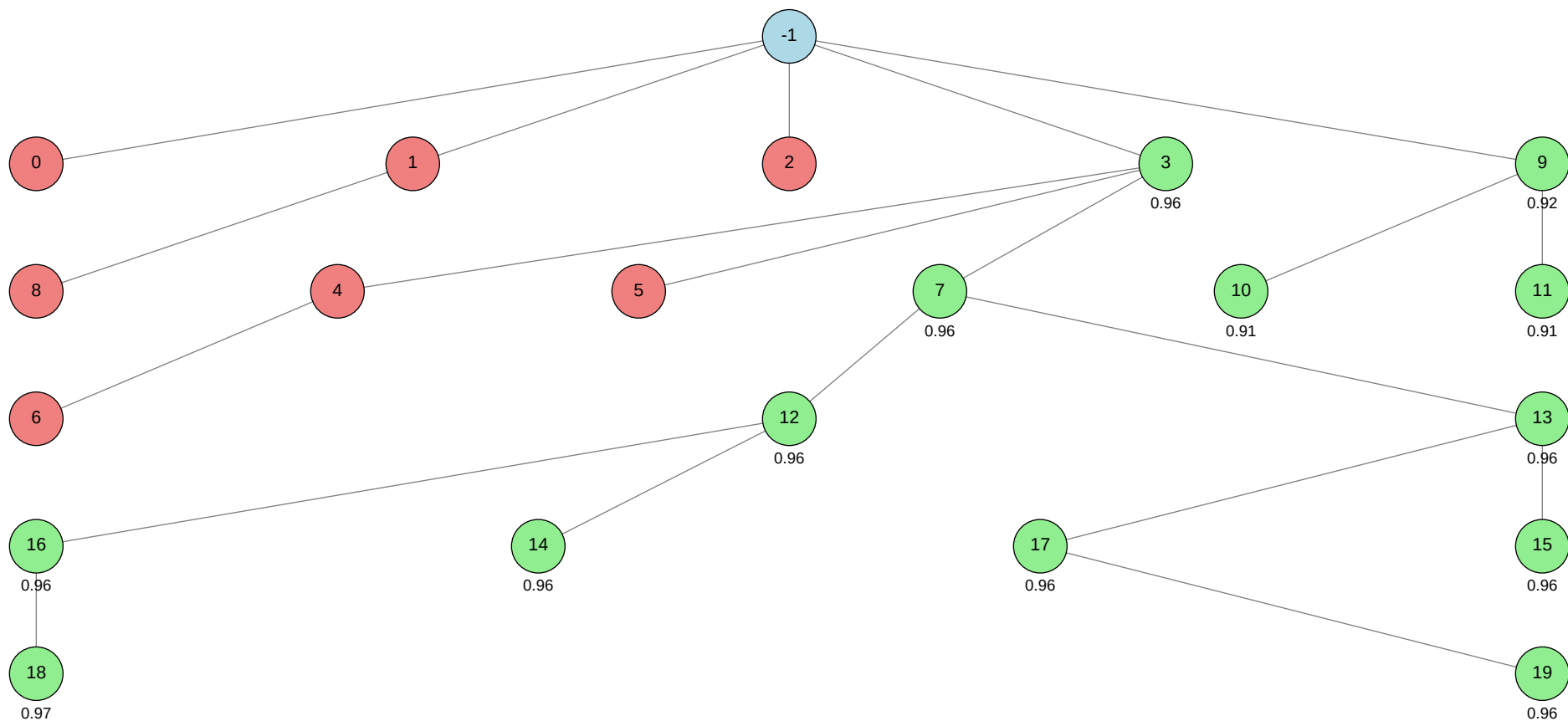
Click on any node in the tree to jump to its detailed information.

Total Nodes: 21 | Best Validation Score: 0.9667 (Node 18)

## Node Tree:

## Table of Contents:

| Node ID                                              | Stage | Status | Tool                 |
|------------------------------------------------------|-------|--------|----------------------|
| <a color="blue" href="#node_1">&lt;-1&lt;/a&gt;</a>  |       |        |                      |
| <a color="blue" href="#node_0">&gt;0&lt;/a&gt;</a>   |       |        | autogluon.multimodal |
| <a color="blue" href="#node_1">&gt;1&lt;/a&gt;</a>   |       |        | machine learning     |
| <a color="blue" href="#node_2">&gt;2&lt;/a&gt;</a>   |       |        | FlagEmbedding        |
| <a color="blue" href="#node_3">&gt;3&lt;/a&gt;</a>   |       |        | autogluon.multimodal |
| <a color="blue" href="#node_4">&gt;4&lt;/a&gt;</a>   |       |        | autogluon.multimodal |
| <a color="blue" href="#node_5">&gt;5&lt;/a&gt;</a>   |       |        | autogluon.multimodal |
| <a color="blue" href="#node_6">&gt;6&lt;/a&gt;</a>   |       |        | autogluon.multimodal |
| <a color="blue" href="#node_7">&gt;7&lt;/a&gt;</a>   |       |        | autogluon.multimodal |
| <a color="blue" href="#node_8">&gt;8&lt;/a&gt;</a>   |       |        | machine learning     |
| <a color="blue" href="#node_9">&gt;9&lt;/a&gt;</a>   |       |        | machine learning     |
| <a color="blue" href="#node_10">&gt;10&lt;/a&gt;</a> |       |        | machine learning     |
| <a color="blue" href="#node_11">&gt;11&lt;/a&gt;</a> |       |        | machine learning     |
| <a color="blue" href="#node_12">&gt;12&lt;/a&gt;</a> |       |        | autogluon.multimodal |
| <a color="blue" href="#node_13">&gt;13&lt;/a&gt;</a> |       |        | autogluon.multimodal |
| <a color="blue" href="#node_14">&gt;14&lt;/a&gt;</a> |       |        | autogluon.multimodal |
| <a color="blue" href="#node_15">&gt;15&lt;/a&gt;</a> |       |        | autogluon.multimodal |
| <a color="blue" href="#node_16">&gt;16&lt;/a&gt;</a> |       |        | autogluon.multimodal |
| <a color="blue" href="#node_17">&gt;17&lt;/a&gt;</a> |       |        | autogluon.multimodal |
| <a color="blue" href="#node_18">&gt;18&lt;/a&gt;</a> |       |        | autogluon.multimodal |
| <a color="blue" href="#node_19">&gt;19&lt;/a&gt;</a> |       |        | autogluon.multimodal |



| Status  | Color |
|---------|-------|
| Success | Green |
| Failure | Red   |
| Neutral | Blue  |

Node Details:

Node -1 (root)

| Property             | Value               |
|----------------------|---------------------|
| uct_value            | 0.3400              |
| ctime                | 2026-05-22 16:55:14 |
| debug_attempts       | 0                   |
| depth                | 0                   |
| execution_time       | 0.0                 |
| expected_child_count | 0                   |
| failure_visits       | 7                   |
| id                   | -1                  |
| is_debug_successful  | ✗                   |
| is_leaf              | ✗                   |
| is_successful        | ✗                   |
| is_terminal          | ✗                   |
| num_children         | 5                   |
| prev_tutorial_prompt | None                |
| stage                | root                |
| time_step            | -1                  |
| tool_used            |                     |
| tools_available      | []                  |
| unvalidated_visits   | 0                   |
| validated_reward     | 12.3422             |
| validated_visits     | 13                  |
| validation_score     | None                |
| visits               | 20                  |
| parent_id            | None                |
| child_ids            | [0, 1, 2, 3, 9]     |



## Node 0 (evolve)

| Property                 | Value                                                         |
|--------------------------|---------------------------------------------------------------|
| uct_value                | 2.4474                                                        |
| ctime                    | 2026-05-22 16:56:21                                           |
| debug_attempts           | 0                                                             |
| depth                    | 1                                                             |
| execution_time           | 0.0                                                           |
| expected_child_count     | 0                                                             |
| exploitation             | 0.0                                                           |
| exploration              | 1.482079962591042                                             |
| failure_penalty          | -0.0                                                          |
| failure_visits           | 1                                                             |
| id                       | 0                                                             |
| is_debug_successful      | ✗                                                             |
| is_leaf                  | ✓                                                             |
| is_successful            | ✗                                                             |
| is_terminal              | ✓                                                             |
| normalized_failure_visit | 0                                                             |
| num_children             | 0                                                             |
| prev_tutorial_prompt     | None                                                          |
| stage                    | evolve                                                        |
| time_step                | 0                                                             |
| tool_used                | autogluon.multimodal                                          |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding'] |
| unvalidated_visits       | 0                                                             |
| validated_contribution   | 0.0                                                           |
| validated_reward         | 0.0                                                           |
| validated_visits         | 0                                                             |
| validation_score         | None                                                          |

|           |    |
|-----------|----|
| visits    | 1  |
| parent_id | -1 |
| child_ids | [] |

### **Error Message:**

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 32-basic\_prompt/node\_0/conda\_env

Resolved 235 packages in 1.39s

Uninstalled 1 package in 1.77s

Installed 233 packages in 1m 13s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0
- + httpx==1.0.dev3
- + huggingface-hub==0.36.2
- + hyperopt==0.2.7

+ idna==3.1... [truncated, 9604 chars total]

### ***Error Analysis:***

ERROR\_SUMMARY: The process was terminated after reaching the 1-hour (3600 seconds) wall-clock time limit, causing model training to be interrupted before completion. As a result, the model may not be fully trained, and subsequent steps such as prediction, saving results, and validation AUROC calculation may not have executed or finished properly.

### **SUGGESTED\_FIX:**

Reduce the training time by setting a lower `time_limit` parameter in `predictor.fit()` (e.g., 3000 seconds or less to allow time for inference and saving), use a smaller model or fewer epochs, or decrease dataset size for faster training. Monitor training progress and ensure that all required steps (prediction, saving, validation) can complete within the 1-hour limit.



## Node 1 (evolve)

| Property                 | Value                                                         |
|--------------------------|---------------------------------------------------------------|
| uct_value                | 1.7306                                                        |
| ctime                    | 2026-05-22 18:01:07                                           |
| debug_attempts           | 1                                                             |
| depth                    | 1                                                             |
| execution_time           | 0.0                                                           |
| expected_child_count     | 0                                                             |
| exploitation             | 0.0                                                           |
| exploration              | 1.1115599719432814                                            |
| failure_penalty          | -0.0                                                          |
| failure_visits           | 2                                                             |
| id                       | 1                                                             |
| is_debug_successful      | ✗                                                             |
| is_leaf                  | ✗                                                             |
| is_successful            | ✗                                                             |
| is_terminal              | ✓                                                             |
| normalized_failure_visit | 0                                                             |
| num_children             | 1                                                             |
| prev_tutorial_prompt     | None                                                          |
| stage                    | evolve                                                        |
| time_step                | 1                                                             |
| tool_used                | machine learning                                              |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding'] |
| unvalidated_visits       | 0                                                             |
| validated_contribution   | 0.0                                                           |
| validated_reward         | 0.0                                                           |
| validated_visits         | 0                                                             |
| validation_score         | None                                                          |

|           |     |
|-----------|-----|
| visits    | 2   |
| parent_id | -1  |
| child_ids | [8] |

### **Error Message:**

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

warning: Requirements file `~/home/anri21/.conda/envs/mlzero\_env/lib/python3.11/site-packages/autogluon/assistant/tools\_registry/machine learning/requirements.txt` does not contain any dependencies

Using Python 3.11.15 environment at: 32-basic\_prompt/node\_1/conda\_env

Resolved 79 packages in 439ms

Uninstalled 1 package in 1.90s

Installed 78 packages in 52.64s

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.1

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ filelock==3.29.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

+ huggingface-hub==1.16.1

+ idna==3.16

+ jinja2==3.1.6

```
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+... [truncated, 4250 chars total]
```

### ***Error Analysis:***

ERROR\_SUMMARY: The script fails with a KeyError: "None of [Index(['skin\_tone', 'alternative\_skin\_tone'], dtype='str')] are in the [columns]" when constructing the test dataset. This occurs because the test.csv file does not contain the columns 'skin\_tone' and 'alternative\_skin\_tone', but the code attempts to access them as tabular features for the test set.

#### SUGGESTED\_FIX:

Update the code to only use tabular features that are present in both train and test datasets. Before defining `tabular_cols`, check which columns exist in both train and test, and use only those for tabular feature extraction and model input; if none exist, set `tabular_cols` to an empty list and adjust the model and dataset accordingly to use image data only for the test set.

## Node 2 (evolve)

| Property                 | Value                                                         |
|--------------------------|---------------------------------------------------------------|
| uct_value                | 2.4474                                                        |
| ctime                    | 2026-05-22 18:07:36                                           |
| debug_attempts           | 0                                                             |
| depth                    | 1                                                             |
| execution_time           | 0.0                                                           |
| expected_child_count     | 0                                                             |
| exploitation             | 0.0                                                           |
| exploration              | 1.2131672923786587                                            |
| failure_penalty          | -0.0                                                          |
| failure_visits           | 1                                                             |
| id                       | 2                                                             |
| is_debug_successful      | ✗                                                             |
| is_leaf                  | ✓                                                             |
| is_successful            | ✗                                                             |
| is_terminal              | ✗                                                             |
| normalized_failure_visit | 0                                                             |
| num_children             | 0                                                             |
| prev_tutorial_prompt     | None                                                          |
| stage                    | evolve                                                        |
| time_step                | 2                                                             |
| tool_used                | FlagEmbedding                                                 |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding'] |
| unvalidated_visits       | 0                                                             |
| validated_contribution   | 0.0                                                           |
| validated_reward         | 0.0                                                           |
| validated_visits         | 0                                                             |
| validation_score         | None                                                          |

|           |    |
|-----------|----|
| visits    | 1  |
| parent_id | -1 |
| child_ids | [] |

### **Error Message:**

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 32-basic\_prompt/node\_2/conda\_env

Resolved 109 packages in 777ms

Uninstalled 1 package in 2.11s

Installed 108 packages in 50.66s

+ accelerate==1.13.0

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiosignal==1.4.0

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ anyio==4.13.0

+ attrs==26.1.0

+ beautifulsoup4==4.14.3

+ cbor==1.0.0

+ certifi==2026.5.20

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ click==8.4.1

+ cuda-bindings==13.2.0

+ cuda-pathfinder==1.5.4

+ cuda-toolkit==13.0.2

+ datasets==4.8.5

+ dill==0.4.1

+ faiss-cpu==1.13.2

+ filelock==3.29.0

+ flagembedding==1.4.0

+ frozenlist==1.8.0

+ fsspec==2026.2.0

+ h11==0.16.0

+ hf-xet==1.5.0

+ httpcore==1.0.9

+ httpx==0.28.1

- + huggingface-hub==1.16.1
- + idna==3.16
- + ijson==3.5.0
- + inscriptis==2.7.1
- + ir-datasets==0.5.11
- + jinja2==3.1.6
- + joblib==1.5.3
- + lxml==6.0.4
- + lz4==4.4.5
- + markdown-it-py==4.2.0
- + markupsafe==3.0.3
- + mdurl==0.1.2
- + mpmath==1.3.0
- + multidict==6.7.1
- + multiprocessing==0.70.19
- + narwhals==2.21.2
- + networkx==3.6.1
- + numpy==2.4.6
- + nvidia-cublas==13.1.1.3
- + nvidia-cuda-cupti==13.0.85
- + nvidia-cuda-nvrtc==13.0.88
- + nvidia-cuda-runtime==13.0.96
- + nvidia-cudnn-cu13==9.20.0.48
- + nvidia-cufft==12.0.0.61
- + nvidia-cufile==1.15.1.6
- + nvidia-curand==10.4.0.35
- + nvidia-cusolver==12.0.4.66
- + nvidia-cusparselt-cu13==0.8.1
- + nvidia-cusparse==12.6.3.3
- + nvidia-cusparse-cu13==0.8.1
- + nvidia-nccl-cu13==2.29.7
- + nvidia-nvjitlink==13.0.88
- + nvidia-nvshmem-cu13==3.4.5
- + nvidia-nvtx==13.0.85
- + opencv-python==4.13.0.92
- + pandas==3.0.3
- + peft==0.19.1
- + pillow==12.2.0
- + propcache==0.5.2
- + protobuf==7.35.0
- + psutil==7.2.2
- + pyarrow==24.0.0
- + pydantic==2.14.0a1
- + pydantic-core==2.47.0
- + pygments==2.20.0
- + python-calamine==0.6.2

+ python-dateutil==2.9.0.post0  
... [truncated, 3451 chars total]

### ***Error Analysis:***

ERROR\_SUMMARY: The error occurs because the script attempts to import ``BGEM3Image`` from the ``FlagEmbedding`` package, but this class does not exist in the installed version of FlagEmbedding. This is likely due to either a typo in the class name, a mismatch between the code and the FlagEmbedding API, or using an outdated or incompatible version of the FlagEmbedding library that does not provide ``BGEM3Image``.

### **SUGGESTED\_FIX:**

Check the official [FlagEmbedding documentation](<https://github.com/FlagOpen/FlagEmbedding>) or the installed package to confirm the correct class name for image embedding (commonly ``FlagModel`` or similar, not ``BGEM3Image``). Replace ``from FlagEmbedding import BGEM3Image`` with the correct import and update the model instantiation accordingly. If image embedding is not supported in your installed version, either upgrade FlagEmbedding or use an alternative image embedding model (e.g., torchvision or timm) for feature extraction.



### Node 3 (debug)

| Property                 | Value                                                         |
|--------------------------|---------------------------------------------------------------|
| uct_value                | 1.3364                                                        |
| avg_raw_score            | 0.9610555555555556                                            |
| ctime                    | 2026-05-22 18:13:37                                           |
| debug_attempts           | 1                                                             |
| depth                    | 2                                                             |
| execution_time           | 0.0                                                           |
| expected_child_count     | 0                                                             |
| exploitation             | 0.638706140350877                                             |
| exploration              | 0.700422462826868                                             |
| failure_penalty          | -0.041666666666666664                                         |
| failure_visits           | 3                                                             |
| id                       | 3                                                             |
| is_debug_successful      | ✗                                                             |
| is_leaf                  | ✗                                                             |
| is_successful            | ✓                                                             |
| is_terminal              | ✗                                                             |
| normalized_failure_visit | 1                                                             |
| normalized_score         | 0.9071637426900583                                            |
| num_children             | 3                                                             |
| prev_tutorial_prompt     | None                                                          |
| stage                    | debug                                                         |
| time_step                | 3                                                             |
| tool_used                | autogluon.multimodal                                          |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding'] |
| unvalidated_visits       | 0                                                             |
| validated_contribution   | 0.6803728070175437                                            |
| validated_reward         | 9.6092                                                        |

|                  |           |
|------------------|-----------|
| validated_visits | 10        |
| validated_weight | 0.75      |
| validation_score | 0.9576    |
| visits           | 13        |
| parent_id        | -1        |
| child_ids        | [4, 5, 7] |

Node 4 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 1.6013                                                                                                                                                                 |
| ctime                    | 2026-05-22 19:10:28                                                                                                                                                    |
| debug_attempts           | 1                                                                                                                                                                      |
| depth                    | 3                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.0                                                                                                                                                                    |
| exploration              | 1.5761206197432238                                                                                                                                                     |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 2                                                                                                                                                                      |
| id                       | 4                                                                                                                                                                      |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✗                                                                                                                                                                      |
| is_successful            | ✗                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| num_children             | 1                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 4                                                                                                                                                                      |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |
| validated_contribution   | 0.0                                                                                                                                                                    |
| validated_reward         | 0.0                                                                                                                                                                    |

|                  |      |
|------------------|------|
| validated_visits | 0    |
| validation_score | None |
| visits           | 2    |
| parent_id        | 3    |
| child_ids        | [6]  |

### **Error Message:**

```
stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
Using Python 3.11.15 environment at: 32-basic_prompt/node_4/conda_env
Resolved 235 packages in 1.12s
Uninstalled 1 package in 2.00s
Installed 233 packages in 1m 15s
+ absl-py==2.4.0
+ accelerate==1.13.0
+ adagio==0.2.6
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiohttp-cors==0.8.1
+ aiosignal==1.4.0
+ alembic==1.18.4
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ antlr4-python3-runtime==4.9.3
+ attrs==26.1.0
+ autogluon==1.5.1b20260522
+ autogluon-common==1.5.1b20260522
+ autogluon-core==1.5.1b20260522
+ autogluon-features==1.5.1b20260522
+ autogluon-multimodal==1.5.1b20260522
+ autogluon-tabular==1.5.1b20260522
+ autogluon-timeseries==1.5.1b20260522
+ beartype==0.22.9
+ beautifulsoup4==4.14.3
+ blis==1.3.3
+ boto3==1.43.13
+ botocore==1.43.13
+ catalogue==2.0.10
```

- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

+ httpx==1.0.dev3  
+ huggingface-hub==0.36.2  
+ hyperopt==0.2.7  
+ idna==3.1... [truncated, 10846 chars total]

### ***Error Analysis:***

ERROR\_SUMMARY: The error occurs because the hyperparameter keys used for optimization in the AutoGluon MultiModalPredictor ("optimization.max\_epochs", "optimization.learning\_rate", and "optimization.lr\_scheduler") are deprecated and no longer recognized in the current version of AutoGluon. The correct keys now use the "optim." prefix (e.g., "optim.max\_epochs", "optim.lr", "optim.lr\_scheduler"), and using the old keys results in a KeyError during model configuration.

### ***SUGGESTED\_FIX:***

Update your `hyperparameters` dictionary in the `predictor.fit()` call to use the new "optim." prefix for all optimization-related keys. Specifically, replace "optimization.max\_epochs" with "optim.max\_epochs", "optimization.learning\_rate" with "optim.lr", and "optimization.lr\_scheduler" with "optim.lr\_scheduler". Double-check all hyperparameter keys for deprecated prefixes and update them accordingly to match the latest AutoGluon documentation.

Node 5 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 2.2646                                                                                                                                                                 |
| ctime                    | 2026-05-22 19:19:00                                                                                                                                                    |
| debug_attempts           | 0                                                                                                                                                                      |
| depth                    | 3                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.0                                                                                                                                                                    |
| exploration              | 1.664857771836881                                                                                                                                                      |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 1                                                                                                                                                                      |
| id                       | 5                                                                                                                                                                      |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✓                                                                                                                                                                      |
| is_successful            | ✗                                                                                                                                                                      |
| is_terminal              | ✓                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| num_children             | 0                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 5                                                                                                                                                                      |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |
| validated_contribution   | 0.0                                                                                                                                                                    |
| validated_reward         | 0.0                                                                                                                                                                    |

|                  |      |
|------------------|------|
| validated_visits | 0    |
| validation_score | None |
| visits           | 1    |
| parent_id        | 3    |
| child_ids        | []   |

### **Error Message:**

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 32-basic\_prompt/node\_5/conda\_env

Resolved 235 packages in 1.11s

Uninstalled 1 package in 1.76s

Installed 233 packages in 55.79s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10



- + catboost==1.2.10
- + certifi==2026.5.20
- + cffi==2.0.0
- + chardet==7.4.3
- + charset-normalizer==3.4.7
- + chronos-forecasting==2.2.2
- + click==8.4.1
- + cloudpathlib==0.24.0
- + cloudpickle==3.1.2
- + colorama==0.4.6
- + colorful==0.6.0a1
- + colorlog==6.10.1
- + confection==1.3.3
- + contourpy==1.3.3
- + coreforecast==0.0.16
- + cryptography==48.0.0
- + cycler==0.12.1
- + cymem==2.0.13
- + datasets==4.0.0
- + defusedxml==0.7.1
- + dill==0.3.8
- + distlib==0.4.0
- + einops==0.8.2
- + einx==0.4.3
- + evaluate==0.4.6
- + fastai==2.8.5
- + fastcore==1.8.18
- + fastdownload==0.0.7
- + fastprogress==1.0.5
- + fasttransform==0.0.2
- + filelock==3.29.0
- + fonttools==4.63.0
- + frozendict==2.4.7
- + frozenlist==1.8.0
- + fsspec==2025.3.0
- + fugue==0.9.7
- + future==1.0.0
- + gdown==6.0.0
- + gluonts==0.16.2
- + google-api-core==2.30.3
- + google-auth==2.53.0
- + googleapis-common-protos==1.75.0
- + graphviz==0.21
- + grpcio==1.81.0rc1
- + hf-xet==1.5.0

+ httpx==1.0.dev3  
+ huggingface-hub==0.36.2  
+ hyperopt==0.2.7  
+ idna==3.1... [truncated, 10024 chars total]

### ***Error Analysis:***

ERROR\_SUMMARY: The error occurs because the hyperparameter "optim.lr\_scheduler" is not a valid or recognized key in the current AutoGluon MultiModal configuration. When the script attempts to set "optim.lr\_scheduler" to "cosine" in the hyperparameters dictionary, AutoGluon raises a KeyError since this key does not exist in its configuration schema.

### **SUGGESTED\_FIX:**

Remove the "optim.lr\_scheduler" entry from the hyperparameters dictionary in your predictor.fit() call. If you wish to set a learning rate schedule, use the correct key—such as "optim.lr\_schedule" (e.g., "cosine\_decay", "polynomial\_decay", or "linear\_decay")—as documented in the AutoGluon tutorials. Double-check the available hyperparameter keys for your AutoGluon version and only use supported options.

Node 6 (debug)

| Property             | Value                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value            | 1.1772                                                                                                                                                                 |
| ctime                | 2026-05-22 19:27:13                                                                                                                                                    |
| debug_attempts       | 1                                                                                                                                                                      |
| depth                | 4                                                                                                                                                                      |
| execution_time       | 0.0                                                                                                                                                                    |
| expected_child_count | 0                                                                                                                                                                      |
| failure_visits       | 1                                                                                                                                                                      |
| id                   | 6                                                                                                                                                                      |
| is_debug_successful  | ✗                                                                                                                                                                      |
| is_leaf              | ✓                                                                                                                                                                      |
| is_successful        | ✗                                                                                                                                                                      |
| is_terminal          | ✗                                                                                                                                                                      |
| num_children         | 0                                                                                                                                                                      |
| prev_tutorial_prompt | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                | debug                                                                                                                                                                  |
| time_step            | 6                                                                                                                                                                      |
| tool_used            | autogluon.multimodal                                                                                                                                                   |
| tools_available      | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits   | 0                                                                                                                                                                      |
| validated_reward     | 0.0                                                                                                                                                                    |
| validated_visits     | 0                                                                                                                                                                      |
| validation_score     | None                                                                                                                                                                   |
| visits               | 1                                                                                                                                                                      |
| parent_id            | 4                                                                                                                                                                      |
| child_ids            | []                                                                                                                                                                     |

## Error Message:

stderr:

==> WARNING: A newer version of conda exists. <==

current version: 25.11.0

latest version: 26.5.0

Please update conda by running

\$ conda update -n base -c conda-forge conda

Using Python 3.11.15 environment at: 32-basic\_prompt/node\_6/conda\_env

Resolved 235 packages in 1.16s

Uninstalled 1 package in 1.83s

Installed 233 packages in 54.16s

+ absl-py==2.4.0

+ accelerate==1.13.0

+ adagio==0.2.6

+ aiohappyeyeballs==2.6.2

+ aiohttp==3.13.5

+ aiohttp-cors==0.8.1

+ aiosignal==1.4.0

+ alembic==1.18.4

+ annotated-doc==0.0.4

+ annotated-types==0.7.0

+ antlr4-python3-runtime==4.9.3

+ attrs==26.1.0

+ autogluon==1.5.1b20260522

+ autogluon-common==1.5.1b20260522

+ autogluon-core==1.5.1b20260522

+ autogluon-features==1.5.1b20260522

+ autogluon-multimodal==1.5.1b20260522

+ autogluon-tabular==1.5.1b20260522

+ autogluon-timeseries==1.5.1b20260522

+ beartype==0.22.9

+ beautifulsoup4==4.14.3

+ blis==1.3.3

+ boto3==1.43.13

+ botocore==1.43.13

+ catalogue==2.0.10

+ catboost==1.2.10

+ certifi==2026.5.20

+ cffi==2.0.0

+ chardet==7.4.3

+ charset-normalizer==3.4.7

+ chronos-forecasting==2.2.2

+ click==8.4.1

+ cloudpathlib==0.24.0

+ cloudpickle==3.1.2  
+ colorama==0.4.6  
+ colorful==0.6.0a1  
+ colorlog==6.10.1  
+ confection==1.3.3  
+ contourpy==1.3.3  
+ coreforecast==0.0.16  
+ cryptography==48.0.0  
+ cycler==0.12.1  
+ cymem==2.0.13  
+ datasets==4.0.0  
+ defusedxml==0.7.1  
+ dill==0.3.8  
+ distlib==0.4.0  
+ einops==0.8.2  
+ einx==0.4.3  
+ evaluate==0.4.6  
+ fastai==2.8.5  
+ fastcore==1.8.18  
+ fastdownload==0.0.7  
+ fastprogress==1.0.5  
+ fasttransform==0.0.2  
+ filelock==3.29.0  
+ fonttools==4.63.0  
+ frozendict==2.4.7  
+ frozenlist==1.8.0  
+ fsspec==2025.3.0  
+ fugue==0.9.7  
+ future==1.0.0  
+ gdown==6.0.0  
+ gluonts==0.16.2  
+ google-api-core==2.30.3  
+ google-auth==2.53.0  
+ googleapis-common-protos==1.75.0  
+ graphviz==0.21  
+ grpcio==1.81.0rc1  
+ hf-xet==1.5.0  
+ httpx==1.0.dev3  
+ huggingface-hub==0.36.2  
+ hyperopt==0.2.7  
+ idna==3.1... [truncated, 9957 chars total]

### ***Error Analysis:***

ERROR\_SUMMARY: The error is a KeyError caused by passing the hyperparameter "fusion.enable" to AutoGluon's MultiModalPredictor.fit(), but this key does not exist in the current configuration schema. As a result, the training process fails during configuration override, preventing model training and prediction.

SUGGESTED\_FIX: Remove the "fusion.enable" key from the hyperparameters dictionary passed to predictor.fit(). Review the official AutoGluon documentation for supported hyperparameters, and only include keys that are valid for your installed version; if you want to enable late fusion or multimodal fusion, use the recommended approach (e.g., by specifying multiple model names in "model.names" or using default fusion behavior).

Node 7 (debug)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 1.6659                                                                                                                                                                 |
| avg_raw_score            | 0.9614875                                                                                                                                                              |
| ctime                    | 2026-05-22 19:35:12                                                                                                                                                    |
| debug_attempts           | 1                                                                                                                                                                      |
| depth                    | 4                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.9142680921052638                                                                                                                                                     |
| exploration              | 0.7880603098716119                                                                                                                                                     |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 0                                                                                                                                                                      |
| id                       | 7                                                                                                                                                                      |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✗                                                                                                                                                                      |
| is_successful            | ✓                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| normalized_score         | 0.9142680921052638                                                                                                                                                     |
| num_children             | 2                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | debug                                                                                                                                                                  |
| time_step                | 7                                                                                                                                                                      |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |

|                        |                    |
|------------------------|--------------------|
| validated_contribution | 0.9142680921052638 |
| validated_reward       | 8.6516             |
| validated_visits       | 9                  |
| validated_weight       | 1.0                |
| validation_score       | 0.9616             |
| visits                 | 9                  |
| parent_id              | 3                  |
| child_ids              | [12, 13]           |



## Node 8 (debug)

| Property             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value            | 1.1772                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ctime                | 2026-05-22 19:56:53                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| debug_attempts       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| depth                | 2                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| execution_time       | 0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| expected_child_count | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| failure_visits       | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| id                   | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_debug_successful  | ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_leaf              | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_successful        | ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_terminal          | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| num_children         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| prev_tutorial_prompt | <p>### Condensed: Intro to Use Machine Learning or Deep Learning algorithms</p> <p># Condensed: Intro to Use Machine Learning or Deep Learning algorithms</p> <p>Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you</p> <p># Intro to Use Machine Learning or Deep Learning algorithms</p> <p>You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them</p> |
| stage                | debug                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
| time_step            | 8                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| tool_used            | machine learning                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| tools_available      | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                                                                                                                                                                                                                                                                                                                         |
| unvalidated_visits   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| validated_reward     | 0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| validated_visits     | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                  |      |
|------------------|------|
| validation_score | None |
| visits           | 1    |
| parent_id        | 1    |
| child_ids        | []   |

### Error Message:

```

stderr:
==> WARNING: A newer version of conda exists. <==
current version: 25.11.0
latest version: 26.5.0
Please update conda by running
$ conda update -n base -c conda-forge conda
warning: Requirements file `~/home/anri21/.conda/envs/mlzero_env/lib/python3.11/site-packages/autogluon/assistant/tools_registry/machine learning/requirements.txt`
does not contain any dependencies
Using Python 3.11.15 environment at: 32-basic_prompt/node_8/conda_env
Resolved 79 packages in 1.05s
Uninstalled 1 package in 2.03s
Installed 78 packages in 43.20s
+ aiohappyeyeballs==2.6.2
+ aiohttp==3.13.5
+ aiosignal==1.4.0
+ annotated-doc==0.0.4
+ annotated-types==0.7.0
+ anyio==4.13.0
+ attrs==26.1.0
+ certifi==2026.5.20
+ chardet==7.4.3
+ charset-normalizer==3.4.7
+ click==8.4.1
+ cuda-bindings==13.2.0
+ cuda-pathfinder==1.5.4
+ cuda-toolkit==13.0.2
+ datasets==4.8.5
+ dill==0.4.1
+ filelock==3.29.0
+ frozenlist==1.8.0
+ fsspec==2026.2.0
+ h11==0.16.0
+ hf-xet==1.5.0
+ httpcore==1.0.9
+ httpx==0.28.1
+ huggingface-hub==1.16.1
+ idna==3.16

```

```
+ jinja2==3.1.6
+ markdown-it-py==4.2.0
+ markupsafe==3.0.3
+ mdurl==0.1.2
+ mpmath==1.3.0
+ multidict==6.7.1
+ multiprocessing==0.70.19
+ networkx==3.6.1
+ numpy==2.4.6
+ nvidia-cublas==13.1.1.3
+ nvidia-cuda-cupti==13.0.85
+ nvidia-cuda-nvrtc==13.0.88
+ nvidia-cuda-runtime==13.0.96
+ nvidia-cudnn-cu13==9.20.0.48
+ nvidia-cufft==12.0.0.61
+ nvidia-cufile==1.15.1.6
+ nvidia-curand==10.4.0.35
+ nvidia-cusolver==12.0.4.66
+ nvidia-cusparselt-cu13==0.8.1
+ nvidia-nccl-cu13==2.29.7
+ nvidia-nvjitlink==13.0.88
+ nvidia-nvshmem-cu13==3.4.5
+ nvidia-nvtx==13.0.85
+ opencv-python==4.13.0.92
+ pandas==3.0.3
+ pillow==12.2.0
+ propcache==0.5.2
+ pyarrow==24.0.0
+ pydantic==2.14.0a1
+ pydantic-core==2.47.0
+ pygments==2.20.0
+ python-calamine==0.6.2
+ python-dateutil==2.9.0.post0
+ pytz==2026.2
+ pyyaml==6.0.3
+ requests==2.34.2
+ rich==15.0.0
- setuptools==82.0.1
+... [truncated, 5181 chars total]
```

### ***Error Analysis:***

ERROR\_SUMMARY: The script attempts to load pretrained weights for the EfficientNetV2-S model using timm, but these weights are not available in the current timm installation or model registry. As a result, a RuntimeError is raised, instructing to use pretrained=False for random initialization.

SUGGESTED\_FIX: Change the LesionNet model initialization in your code to use `pretrained=False` when calling `timm.create_model` for "efficientnetv2\_s". This will initialize the model with random weights and allow training to proceed, though you may want to consider using a different backbone with available pretrained weights for better performance.

## Node 9 (debug)

| Property                 | Value                                                         |
|--------------------------|---------------------------------------------------------------|
| uct_value                | 1.4969                                                        |
| avg_raw_score            | 0.911                                                         |
| ctime                    | 2026-05-22 20:03:07                                           |
| debug_attempts           | 2                                                             |
| depth                    | 2                                                             |
| execution_time           | 0.0                                                           |
| expected_child_count     | 0                                                             |
| exploitation             | 0.08388157894736836                                           |
| exploration              | 1.050633694240302                                             |
| failure_penalty          | -0.0                                                          |
| failure_visits           | 0                                                             |
| id                       | 9                                                             |
| is_debug_successful      | ✗                                                             |
| is_leaf                  | ✗                                                             |
| is_successful            | ✓                                                             |
| is_terminal              | ✗                                                             |
| normalized_failure_visit | 0                                                             |
| normalized_score         | 0.08388157894736836                                           |
| num_children             | 2                                                             |
| prev_tutorial_prompt     | None                                                          |
| stage                    | debug                                                         |
| time_step                | 9                                                             |
| tool_used                | machine learning                                              |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding'] |
| unvalidated_visits       | 0                                                             |
| validated_contribution   | 0.08388157894736836                                           |
| validated_reward         | 2.733                                                         |

|                  |          |
|------------------|----------|
| validated_visits | 3        |
| validated_weight | 1.0      |
| validation_score | 0.9212   |
| visits           | 3        |
| parent_id        | -1       |
| child_ids        | [10, 11] |

## Node 10 (evolve)

| Property             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value            | 1.4821                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ctime                | 2026-05-22 20:49:32                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| debug_attempts       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| depth                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| execution_time       | 0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| expected_child_count | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| failure_visits       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| id                   | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| is_debug_successful  | ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_leaf              | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_successful        | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_terminal          | ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| num_children         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| prev_tutorial_prompt | <p>### Condensed: Intro to Use Machine Learning or Deep Learning algorithms</p> <p># Condensed: Intro to Use Machine Learning or Deep Learning algorithms</p> <p>Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you</p> <p># Intro to Use Machine Learning or Deep Learning algorithms</p> <p>You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them</p> |
| stage                | evolve                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| time_step            | 10                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| tool_used            | machine learning                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| tools_available      | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                                                                                                                                                                                                                                                                                                                         |
| unvalidated_visits   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| validated_reward     | 0.9059                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| validated_visits     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                  |        |
|------------------|--------|
| validation_score | 0.9059 |
| visits           | 1      |
| parent_id        | 9      |
| child_ids        | []     |



## Node 11 (evolve)

| Property             | Value                                                                                                                                                                                                                                                                                                                                                                                                                                                                 |
|----------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value            | 1.4821                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| ctime                | 2026-05-22 21:49:53                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| debug_attempts       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| depth                | 3                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| execution_time       | 0.0                                                                                                                                                                                                                                                                                                                                                                                                                                                                   |
| expected_child_count | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| failure_visits       | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| id                   | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| is_debug_successful  | ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_leaf              | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_successful        | ✓                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| is_terminal          | ✗                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| num_children         | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| prev_tutorial_prompt | <p>### Condensed: Intro to Use Machine Learning or Deep Learning algorithms</p> <p># Condensed: Intro to Use Machine Learning or Deep Learning algorithms</p> <p>Summary: You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you</p> <p># Intro to Use Machine Learning or Deep Learning algorithms</p> <p>You can use any machine learning or deep learning algorithms to complete the tasks. Make sure you have them</p> |
| stage                | evolve                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| time_step            | 11                                                                                                                                                                                                                                                                                                                                                                                                                                                                    |
| tool_used            | machine learning                                                                                                                                                                                                                                                                                                                                                                                                                                                      |
| tools_available      | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                                                                                                                                                                                                                                                                                                                         |
| unvalidated_visits   | 0                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
| validated_reward     | 0.9059                                                                                                                                                                                                                                                                                                                                                                                                                                                                |
| validated_visits     | 1                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |

|                  |        |
|------------------|--------|
| validation_score | 0.9059 |
| visits           | 1      |
| parent_id        | 9      |
| child_ids        | []     |

Node 12 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 1.9797                                                                                                                                                                 |
| avg_raw_score            | 0.9625499999999999                                                                                                                                                     |
| ctime                    | 2026-05-22 22:51:08                                                                                                                                                    |
| debug_attempts           | 0                                                                                                                                                                      |
| depth                    | 5                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.93174342105263                                                                                                                                                       |
| exploration              | 1.0195130088268243                                                                                                                                                     |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 0                                                                                                                                                                      |
| id                       | 12                                                                                                                                                                     |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✗                                                                                                                                                                      |
| is_successful            | ✓                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| normalized_score         | 0.93174342105263                                                                                                                                                       |
| num_children             | 2                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 12                                                                                                                                                                     |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |

|                        |                    |
|------------------------|--------------------|
| validated_contribution | 0.93174342105263   |
| validated_reward       | 3.8501999999999996 |
| validated_visits       | 4                  |
| validated_weight       | 1.0                |
| validation_score       | 0.9590             |
| visits                 | 4                  |
| parent_id              | 7                  |
| child_ids              | [16, 14]           |

Node 13 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 1.9370                                                                                                                                                                 |
| avg_raw_score            | 0.9600333333333334                                                                                                                                                     |
| ctime                    | 2026-05-22 23:15:52                                                                                                                                                    |
| debug_attempts           | 0                                                                                                                                                                      |
| depth                    | 5                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.8903508771929836                                                                                                                                                     |
| exploration              | 1.1772322201769845                                                                                                                                                     |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 0                                                                                                                                                                      |
| id                       | 13                                                                                                                                                                     |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✗                                                                                                                                                                      |
| is_successful            | ✓                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| normalized_score         | 0.8903508771929836                                                                                                                                                     |
| num_children             | 2                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 13                                                                                                                                                                     |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |

|                        |                    |
|------------------------|--------------------|
| validated_contribution | 0.8903508771929836 |
| validated_reward       | 3.8398000000000003 |
| validated_visits       | 4                  |
| validated_weight       | 1.0                |
| validation_score       | 0.9590             |
| visits                 | 4                  |
| parent_id              | 7                  |
| child_ids              | [17, 15]           |

Node 14 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 2.5530                                                                                                                                                                 |
| avg_raw_score            | 0.9599                                                                                                                                                                 |
| ctime                    | 2026-05-22 23:39:19                                                                                                                                                    |
| debug_attempts           | 0                                                                                                                                                                      |
| depth                    | 6                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.9199318568994882                                                                                                                                                     |
| exploration              | 1.482079962591042                                                                                                                                                      |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 0                                                                                                                                                                      |
| id                       | 14                                                                                                                                                                     |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✓                                                                                                                                                                      |
| is_successful            | ✓                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| normalized_score         | 0.9199318568994882                                                                                                                                                     |
| num_children             | 0                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 14                                                                                                                                                                     |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |

|                        |                    |
|------------------------|--------------------|
| validated_contribution | 0.9199318568994882 |
| validated_reward       | 0.9599             |
| validated_visits       | 1                  |
| validated_weight       | 1.0                |
| validation_score       | 0.9599             |
| visits                 | 1                  |
| parent_id              | 12                 |
| child_ids              | []                 |



Node 15 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 2.5300                                                                                                                                                                 |
| avg_raw_score            | 0.9585                                                                                                                                                                 |
| ctime                    | 2026-05-23 00:05:59                                                                                                                                                    |
| debug_attempts           | 0                                                                                                                                                                      |
| depth                    | 6                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.8651315789473686                                                                                                                                                     |
| exploration              | 1.482079962591042                                                                                                                                                      |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 0                                                                                                                                                                      |
| id                       | 15                                                                                                                                                                     |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✓                                                                                                                                                                      |
| is_successful            | ✓                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| normalized_score         | 0.8651315789473686                                                                                                                                                     |
| num_children             | 0                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 15                                                                                                                                                                     |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |

|                        |                    |
|------------------------|--------------------|
| validated_contribution | 0.8651315789473686 |
| validated_reward       | 0.9585             |
| validated_visits       | 1                  |
| validated_weight       | 1.0                |
| validation_score       | 0.9585             |
| visits                 | 1                  |
| parent_id              | 13                 |
| child_ids              | []                 |

Node 16 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 2.1600                                                                                                                                                                 |
| avg_raw_score            | 0.9646                                                                                                                                                                 |
| ctime                    | 2026-05-23 00:33:13                                                                                                                                                    |
| debug_attempts           | 0                                                                                                                                                                      |
| depth                    | 6                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 1.0                                                                                                                                                                    |
| exploration              | 1.482079962591042                                                                                                                                                      |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 0                                                                                                                                                                      |
| id                       | 16                                                                                                                                                                     |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✗                                                                                                                                                                      |
| is_successful            | ✓                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| normalized_score         | 1.0                                                                                                                                                                    |
| num_children             | 1                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 16                                                                                                                                                                     |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |

|                        |        |
|------------------------|--------|
| validated_contribution | 1.0    |
| validated_reward       | 1.9313 |
| validated_visits       | 2      |
| validated_weight       | 1.0    |
| validation_score       | 0.9646 |
| visits                 | 2      |
| parent_id              | 12     |
| child_ids              | [18]   |

Node 17 (evolve)

| Property                 | Value                                                                                                                                                                  |
|--------------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value                | 2.0859                                                                                                                                                                 |
| avg_raw_score            | 0.9626                                                                                                                                                                 |
| ctime                    | 2026-05-23 01:00:14                                                                                                                                                    |
| debug_attempts           | 0                                                                                                                                                                      |
| depth                    | 6                                                                                                                                                                      |
| execution_time           | 0.0                                                                                                                                                                    |
| expected_child_count     | 0                                                                                                                                                                      |
| exploitation             | 0.9325657894736843                                                                                                                                                     |
| exploration              | 1.482079962591042                                                                                                                                                      |
| failure_penalty          | -0.0                                                                                                                                                                   |
| failure_visits           | 0                                                                                                                                                                      |
| id                       | 17                                                                                                                                                                     |
| is_debug_successful      | ✗                                                                                                                                                                      |
| is_leaf                  | ✗                                                                                                                                                                      |
| is_successful            | ✓                                                                                                                                                                      |
| is_terminal              | ✗                                                                                                                                                                      |
| normalized_failure_visit | 0                                                                                                                                                                      |
| normalized_score         | 0.9325657894736843                                                                                                                                                     |
| num_children             | 1                                                                                                                                                                      |
| prev_tutorial_prompt     | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                    | evolve                                                                                                                                                                 |
| time_step                | 17                                                                                                                                                                     |
| tool_used                | autogluon.multimodal                                                                                                                                                   |
| tools_available          | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits       | 0                                                                                                                                                                      |

|                        |                    |
|------------------------|--------------------|
| validated_contribution | 0.9325657894736843 |
| validated_reward       | 1.9223             |
| validated_visits       | 2                  |
| validated_weight       | 1.0                |
| validation_score       | 0.9626             |
| visits                 | 2                  |
| parent_id              | 13                 |
| child_ids              | [19]               |

Node 18 (evolve)

| Property             | Value                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value            | 2.1772                                                                                                                                                                 |
| ctime                | 2026-05-23 01:19:12                                                                                                                                                    |
| debug_attempts       | 0                                                                                                                                                                      |
| depth                | 7                                                                                                                                                                      |
| execution_time       | 0.0                                                                                                                                                                    |
| expected_child_count | 0                                                                                                                                                                      |
| failure_visits       | 0                                                                                                                                                                      |
| id                   | 18                                                                                                                                                                     |
| is_debug_successful  | ✗                                                                                                                                                                      |
| is_leaf              | ✓                                                                                                                                                                      |
| is_successful        | ✓                                                                                                                                                                      |
| is_terminal          | ✗                                                                                                                                                                      |
| num_children         | 0                                                                                                                                                                      |
| prev_tutorial_prompt | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                | evolve                                                                                                                                                                 |
| time_step            | 18                                                                                                                                                                     |
| tool_used            | autogluon.multimodal                                                                                                                                                   |
| tools_available      | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits   | 0                                                                                                                                                                      |
| validated_reward     | 0.9667                                                                                                                                                                 |
| validated_visits     | 1                                                                                                                                                                      |
| validation_score     | 0.9667                                                                                                                                                                 |
| visits               | 1                                                                                                                                                                      |
| parent_id            | 16                                                                                                                                                                     |
| child_ids            | []                                                                                                                                                                     |





Node 19 (evolve)

| Property             | Value                                                                                                                                                                  |
|----------------------|------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| uct_value            | 2.0621                                                                                                                                                                 |
| ctime                | 2026-05-23 01:42:21                                                                                                                                                    |
| debug_attempts       | 0                                                                                                                                                                      |
| depth                | 7                                                                                                                                                                      |
| execution_time       | 0.0                                                                                                                                                                    |
| expected_child_count | 0                                                                                                                                                                      |
| failure_visits       | 0                                                                                                                                                                      |
| id                   | 19                                                                                                                                                                     |
| is_debug_successful  | ✗                                                                                                                                                                      |
| is_leaf              | ✓                                                                                                                                                                      |
| is_successful        | ✓                                                                                                                                                                      |
| is_terminal          | ✗                                                                                                                                                                      |
| num_children         | 0                                                                                                                                                                      |
| prev_tutorial_prompt | ### Condensed: ```python<br># Condensed: ```python<br><br>Summary: This tutorial demonstrates image classification using AutoGluon MultiModal, covering implementation |
| stage                | evolve                                                                                                                                                                 |
| time_step            | 19                                                                                                                                                                     |
| tool_used            | autogluon.multimodal                                                                                                                                                   |
| tools_available      | ['autogluon.multimodal', 'machine learning', 'FlagEmbedding']                                                                                                          |
| unvalidated_visits   | 0                                                                                                                                                                      |
| validated_reward     | 0.9597                                                                                                                                                                 |
| validated_visits     | 1                                                                                                                                                                      |
| validation_score     | 0.9597                                                                                                                                                                 |
| visits               | 1                                                                                                                                                                      |
| parent_id            | 17                                                                                                                                                                     |
| child_ids            | []                                                                                                                                                                     |

